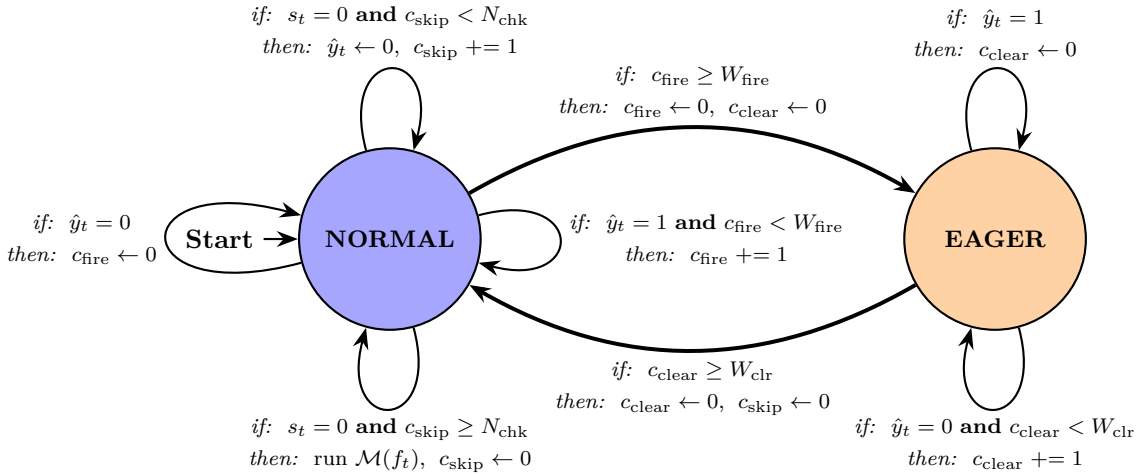




| Symbol                                             | Description                                                                                                                                                            |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <span style="color: blue;">●</span> <b>NORMAL</b>  | Skip module $\mathcal{S}$ is <b>active</b> : static frames are skipped; $\mathcal{M}$ is invoked only when motion is detected or $c_{\text{skip}} \geq N_{\text{chk}}$ |
| <span style="color: orange;">●</span> <b>EAGER</b> | Skip module $\mathcal{S}$ is <b>bypassed</b> : $\mathcal{M}$ runs on <i>every</i> frame to prevent missed detections                                                   |
| $s_t \in \{0, 1\}$                                 | Skip decision from $\mathcal{S}$ : $s_t=0$ no motion detected (skip inference); $s_t=1$ motion detected (run inference)                                                |
| $\hat{y}_t \in \{0, 1\}$                           | Prediction from $\mathcal{M}$ : $\hat{y}_t=1$ fire/smoke; $\hat{y}_t=0$ negative                                                                                       |
| $c_{\text{skip}}$                                  | Consecutive skipped frames (reset to 0 on any inference call)                                                                                                          |
| $c_{\text{fire}}$                                  | Consecutive fire/smoke predictions in NORMAL mode (reset to 0 on any negative)                                                                                         |
| $c_{\text{clear}}$                                 | Consecutive negative predictions in EAGER mode                                                                                                                         |
| $W_{\text{fire}}$                                  | Fire streak length required to enter EAGER mode                                                                                                                        |
| $W_{\text{clr}}$                                   | Clear streak length required to exit EAGER mode                                                                                                                        |
| $N_{\text{chk}}$                                   | Maximum consecutive skips before a forced inference check                                                                                                              |